are derailed—the Justice Department blocks the merger on antitrust grounds, due diligence uncovers skeletons in the acquired company's closet, or the deal is scuttled at the last minute by an argument over who gets to be the new boss, and so on. When mergers fail, prices plummet. Thus, merger arb ekes out small returns most of the time, and then with a small probability it loses big—exactly the pattern we see in Figure 17.7. Mitchell and Pulvino also demonstrate that the rare, large losses arising when mergers fail are correlated with the market (so merger arb has *systematic risk*): in a market-wide crash, many mergers fail and merger activity dries up.

Relative Value

A comprehensive study of fixed income relative value strategies was done by Duarte, Longstaff, and Yu (2005). They examine several "arbitrages," including swap spread arbitrage, yield curve arbitrage, mortgage arbitrage, volatility arbitrage (the differential pricing of risk in different securities markets), and capital structure arbitrage (trading different securities issued by the same firm). All these strategies generate high returns most of the time and then the occasional large loss. Convertible arbitrage also falls under the relative value category as it profits from differential pricing of risk across two securities markets: convertible bonds and equities. This strategy is also vulnerable to extreme market events.[38]

Losses in relative value strategies can be nasty because fixed income HFs usually employ high leverage. Sometimes HFs are blind to the possibility of large losses—as was the case at LTCM when credit spreads widened on news of Russia's default in 1998. Most of the time housing prices are fairly stable and increase over time. But sometimes they don't, like in 2007, when mortgage spreads (over U.S. Treasuries) shot up, causing painful losses for mortgage arb strategies.

Fung and Hsieh (2002) were prescient in their study of fixed income hedge fund returns using data up to 2001. They back-tested the effects of Great Depression-scale shocks to credit spreads and forecast large potential declines saying that "there exists cyclical exposure to risk factors inherent in most Fixed-Income Arbitrage funds that may be masked by the short existence of the funds themselves." That is, a positive and significant historical alpha resulted from the risk of rare events. They were spot on as that rare event arrived toward the end of the 2000s. As Asness said in his definition of HFs, "Every three or four years they deliver a one-in-a-hundred year flood."

HF Industry

Not surprisingly, if most individual HF styles are short volatility, then the entire HF industry is just a short put as well. Jurek and Stafford (2012) show that a simple strategy of selling out-of-the-money puts accurately matches the risk profile of

[38] See Choi, Getmansky, and Tookes (2009) and Agarwal et al. (2011).